Confidential

MINISTRY OF EDUCATION

Katete DISTRICT

Senior Secondary mock examination

Grade 12 – 2024

Design and technology 6045/1

Paper 1 (Material and Design Process)

Marking scheme

Section A.

1. a) i. Name: Try square

ii. 1. Do not use it as a hammer

2. Use the tool for its right purpose at a right time

iii. 1. For testing the flatness of a work piece

2. For checking the squareness of a work piece

b) 1. It's durable

2. It has qood resistance

2. a) i. Junior hacksaw

ii. It is used for cutting thin metals

b) i.

- They don't slipup
- Gears can handle high speed

ii. - In a car engine

- In watches

3. a) i. Single dovetail joint

ii. – Dovetail saw (the learner to sketch)

4. Bulb - Light energy

Speaker - Sound energy

Motor - Kinetic energy

5. a) **Property (any one)**

i. Brass – Malleability, conductivity, corrosion resistance

Use (any one)

- For making musical instruments, for Plumbing, for decorative applications e.g. doors handles, lighting fixtures.

Property (any one)

i. Bronze – Corrosion resistance, Ductile, Brittle

Use (any one)

- For making bearing, making musical instruments, Automobile parts e.g pumps and bells.

b. Movement – Hinges
Security - Locks

Section B [80 marks]
Part I Compulsory

6. a.

i. A situation should have three items (Activity, Concern and place)

ii. Problem should come from the concern in the situation be addressed.

b. i. Design and make an artefact that would be used by…… (Emphasis should be on **user**, **function** and **place**)

ii Possible solution should include the following aspect:

- three dimensions drawing
- proportions
- annotations
- creativity
- good draughtmanship
- Neatness

c. (Any three)

- Material
- Function
- Safety
- Methods of construction
- Size

d. (Any four)

- **Where** the artifact should be used
- **Who** are the users
- **When** will it be used?
- **What** material should be used
- **Why** should it be made
- **How** should it be used

7. a .

i. A. cross filing
B draw filing

ii. Is a concept or plan for a new venture, product or services that has the potential to generate revenue

iii. – construction
- Furniture making
- Metal fabrication
- Electrical installation/ electrician

iv. Mass production is producing in large quantities while it is producing goods within a time frame.

v. These are raw materials that are produced from natural resources and extracted from pits and quarries, including gravel, crushed stones and sand.
Coarse
- Gravel
- Crashed stones
- Liver gravel
Fine
- Liver sand
- Building sand

b .

i. To sketch the Mortice and Tenon joints [exploded views, neatness and proportionality to be considered]

ii. A. G cramp
B. Bench hook / sawing board
C. Quick release cramp

8. **a.** -to prevent the work piece from being damaged

-to prevent drill bit from being damaged

b.

i. To sketch slot X [neatness and proportionality to be considered]

ii. –Wood

-metal

- plastic

c. i. [any three]

- solar
- wind
- hydropower
- geothermal

ii. - sketch the following

1. Reciprocating
2. Oscillating
3. Rotary

9. a. learners to show using sketches and notes on how the base and support [screws, nails, bolt and nut]

b. learner to complete the stopped housing drawing

c.

(a) English bond

(b) -To prevent structure failure

-To ensure stability and strength

- For load distribution

(c) Jointing is finishing the mortar joints between newly laid bricks while pointing is applying a small amount of mortar into the face joint between the bricks

(d) -To ensure that it is straight

- To ensure that it is square

To ensure smoothness

(e) –[any three]

- Pointing trowel
- Brick hammer
- Tuck pointer
- Brick jointer
- Joint raker
- Bucket trowel

(f)

A. Transistor
B. Capacitor

b.

Input

- Microphone
- LDR

Out put

- Motor
-Loud speaker
- Diode

Process

- Transistor

- .